I PUC LBA-1 MODEL PAPER

SUBJECT : CHEMISTRY

UNIT-1: **SOME BASIC CONCEPTS OF CHEMISTRY**

Max : 30Marks

Instructions

1. Question paper has FIVE parts. All parts are compulsory.
2. a. Part-A carries 10 marks. Each question carries 1 mark.
b. Part-B carries 6 marks. Each question carries 2 marks.
c. Part-C carries 06 marks. Each question carries 3 marks.
d. Part-D carries 08marks. Each question carries 4 marks.
3. In Part-A questions, first attempted answer will be considered for awarding marks.
4. Write balanced chemical equations and draw neat labeled diagrams and graphs wherever necessary.
5. Direct answers to the numerical problems without detailed steps and specific unit for final answer will not carry any marks.
6. Use log tables and simple calculator if necessary (use of scientific calculator is not allowed).

PART-A

I. Select the correct option from the given choices. **7 x 1= 7**

1. A mole of any substance contains
   a. 6.022×10^{26} particles
   b. 6.022×10^{22} particles
   c. 6.022×10^{23} particles
   d. 3.022×10^{22} particles

2. In a particular reaction, one of the reactants limits the number of products formed. That is called as
   a. Limiting reagent
   b. Limiting product
   c. Excessive reagent
   d. Excessive reactant.

3. Which of the following statements about a compound is incorrect?
   a. A molecule of a compound has atoms of different elements
   b. A compound cannot be separated into its constituent elements by physical methods of separation.
   c. A compound retains the physical properties of its constituent elements.
   d. The ratio of atoms of atoms of different elements in a compound is fixed.

4. Dalton's atomic theory couldn't explain which of the following?
   a. Law of multiple proportion
   b. law of gaseous volume
   c. law of conservation of mass
   d. law of constant proportion

5. One part of element A combines with 2 parts of another element B. Six parts of element C combines with 4 parts of element B. If A and C combine together the ratio of their weights will be governed by
   a. Law of definite proportion
   b. Law of multiple proportion
   c. Law of reciprocal proportion
   d. Law of conservation of mass.

6. Student performs a titration with different burettes and finds titre values of 25.2 mL, 25.25 mL and 25.0 mL, the number of significant figures in the average titre value is
   a. 1 b. 2 c. 3 d. 4

7. Which of the following is not a pure substance?
   a. Copper
   b. Gold
   c. Sugar solution
   d. Water

II. Fill in the blanks by choosing the appropriate word from those given in the brackets:

[AIDS , HO, kg Cancer ,] **3 x 1 = 3**

8. The S.I unit of mass is _____
9. AZT is used for victims of _____
10. The empirical formula for hydrogen peroxide is_____

PART-B

III. Answer any THREE of the following questions: **3 x 2= 6**

11. What is homogeneous mixture. Give an example.
12. State law of conservation of mass. Illustrate with an example.
13. Name two green house gases?
14. Convert the following
    i) 500Mg into Kg
    ii) 41Pm into metre.
15. Write the significant figures in the following
    a) 2.004 b) 0.0064
16. Write the prefixes for the following
    a) 10^{-18} b) 10^{21}

PART-C

IV. Answer any TWO of the following questions: **2 x 3= 6**

17. Classify the following substances into element, compound and mixture:
    i) milk ii) Dry ice iii) Diamond
18. Write any three postulates of Dalton's atomic theory.
19. What is mole fraction? Write its mathematical expression for a substance 'B' dissolve in substance 'A' and their number of moles are n_B and n_A respectively.
20. Define molality? How it varies with temperature? Justify your answer.

PART-D (PROBLEMS)

V. Answer any TWO of the following questions: **2 x 4= 8**

21. A compound contains 4.07% hydrogen, 24.27% carbon and 71.65% chlorine. Its molar mass is 98.96 g. What are its empirical and molecular formulas?
22. Calculate
    i) the amount of water (g) produced by the combustion of 16 g of methane.
    ii) the molarity of NaOH in the solution prepared by dissolving its 4 g in enough water to form 250 mL of the solution.
23. In the reaction $A + B_2 \rightarrow AB_2$, Identify the limiting reagent, if any, in the following mixtures
    (i) 300 atoms of A + 200 molecules of B
    (ii) 2 mol A + 3 mol B

BLUE PRINT OF LBA -1 MODEL QUESTION PAPER:

UNIT: SOME BASIC CONCEPTS OF CHEMISTRY

UNIT NUMBER	UNIT NAME	TEACHING HOURS	NUMBER OF MARKS	PART-A	PART-B	PART-C	PART-D
				1 MARK	2 MARK	3 MARK	3 MARK
1	SOME BASIC CONCEPTS OF CHEMISTRY	14	30	**10**	**6**	**6**	**8**
TOTAL	NUMBER OF QUESTION			**10/10**	**3/6**	**2/4**	**2/4**
	NUMBER OF MARKS			**10/10**	**6/12**	**6/12**	**8/12**